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DEEP LEARNING AND ITS APPLICATIONS
(CO3133)

Assignment 2 Report Outline

Vietnamese Food Classification with CNN and ResNet-50 on 30VNFoods

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Abstract

Write the abstract after the main text. Cover the single-label 30VNFoods task, duplicate-aware cleaning, the scratch CNN and two ResNet-50 variants, and evaluation on the cleaned test split. State the paired fine-tuning gain, its training cost, and one error pattern supported by the analysis.

Keywords. Vietnamese food classification, 30VNFoods, transfer learning, ResNet-50, error analysis.

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Part I
Problem and Data Description

1 Problem Formulation

Motivate Vietnamese dish recognition and map one RGB image in $\mathbb{R}^{3\times224\times224}$ to 30 logits and a top-1 label. Describe visually similar dishes, hidden ingredients, and background variation. Ask how much adapting the ImageNet backbone improves test accuracy and macro-F1, and what it costs relative to the frozen backbone. Include the scratch CNN as the baseline.

2 Dataset Description

Source and proposal. *Describe 30VNFoods (Kaggle version 11), 25,136 original images, 30 folder labels, web collection, license, storage size, formats, and source bias. Record the confirmed proposal approval and conditions. The cleaned training set has 17,164 images across 30 classes, above the handbook's minimums of 5,000 images and 5 classes. Report minimum per-class support.*

EDA. *Plot class counts, image sizes and aspect ratios. Report the approximately* $3.4\times$ *imbalance and brightness overlap. Tie the observations to macro-F1 and the H1–H4 error hypotheses.*

Source material. `reports/a2/proposal.md`, `reports/a2/approval.md`, `notebooks/a2_eda.ipynb`, and the plots under `results/a2/figures/`.

3 Data Preparation

Split and leakage control. *Compare official and cleaned counts. The cleaned split retains 17,164 training, 2,481 validation, and 4,984 test images after 507 removals. Explain stratification, duplicate clusters as the leakage-control unit, MD5 and perceptual hashes, removal rules for conflicting labels, and the fixed manifest.*

Transforms and example batch. *Contrast training augmentation with evaluation resize and center crop. Document train-set normalization for CNN and ImageNet normalization for ResNet-50. Show an example batch with input and label shapes.*

Source material. `results/a2/splits/split_v2.json`, `results/a2/tables/a2_drop_list.csv`, `src/a2/data.py`, `results/a2/figures/a2_example_batch.png`.

Part II
Methodology

4 Overall Pipeline

Draw the flow from dataset and split manifest through preprocessing, data loader, model, cross-entropy, and optimization. Show the inference path from saved checkpoint to logits, predictions, and evaluation. Label the validation checkpoint decision and the final test evaluation.

Source material. `src/a2/data.py`, `src/a2/trainer.py`, `src/a2/evaluate.py`.

5 Models

5.1 CNN baseline

Diagram the self-implemented four-block CNN: Conv–BatchNorm–ReLU–pool, then global average pooling, dropout, and a 30-way linear head. Annotate intermediate shapes and the 397,086 trainable parameters.

5.2 ImageNet-pretrained ResNet-50

Cite the torchvision backbone and `IMAGENET1K_V2` *weights. Explain the 30-way head and compare head-only training (61,470 trainable parameters) with full fine-tuning (23,569,502). State how gradients and BatchNorm statistics are handled in each arm.*

Source material. `src/a2/models.py`, `src/a2/trainer.py`, saved per-run configurations.

6 Loss and Optimization

Define multiclass cross-entropy on logits. Record Adam, weight decay, scheduler status, and regularization. Tabulate the actual head learning rate (10^{-3} *for both ResNet arms) and backbone learning rate (0 for frozen,* 10^{-4} *for full). Use the saved CNN run configuration for its optimizer settings.*

Source material. `configs/a2.yaml`, `results/a2/runs/`.

7 Training and Evaluation Protocol

Tabulate saved run settings, including hardware/software, precision, batch size, epoch cap, patience, and seeds. The CNN draft used seed 42, batch 64, and a 20-epoch cap. Both ResNet arms used seeds 42–44, batch 32, a 15-epoch cap, and patience 5. See Section 13 for the controlled design.

Define accuracy and macro-F1, with macro-F1 primary. Explain best-validation-macro-F1 checkpoint selection, validation-based illustrative run selection, the final cleaned test protocol, and secondary metrics.

Source material. `results/a2/tables/a2_test_protocol.json`, per-run `config.yaml` and `metrics.json`.

Part III
Implementation Results

8 Training Behaviour

Plot training/validation loss and validation macro-F1. Mark the checkpoint epoch and the last epoch separately. Full seeds 42, 43, and 44 ran 12, 14, and 15 epochs under the same cap. Explain early stopping in the first two runs. Discuss fit, convergence, and implementation issues documented in the logs.

Source material. Per-run `history.csv`, `metrics.json`, and `figures/` under `results/a2/runs/`.

9 Quantitative Results

9.1 Cleaned test performance

Table 1: Cleaned test results over saved checkpoints. Values after ± are sample standard deviations across seeds.

Model	Runs	Accuracy	Macro-F1
CNN baseline	1	0.5451	0.5403
ResNet-50 frozen	3	0.7679 ± 0.0013	0.7720 ± 0.0022
ResNet-50 full	3	0.8908 ± 0.0064	0.8922 ± 0.0063

Report the 4,984-image, 30-class support and note the CNN is a single draft run. Add per-class F1, representative confusion matrices, and validation metrics where they help explain checkpoint selection.

Source material. `results/a2/tables/a2_test_summary.json`, `results/a2/tables/a2_test_evaluation.csv`.

9.2 Compute cost

Compare total and trainable parameters, time per training run, peak training memory where measured, and forward time per image. State the RTX 4060 Laptop training device and Apple MPS evaluation device, batch size, PyTorch version, and timing boundaries.

Source material. `results/a2/tables/a2_test_compute_cost.csv`.

10 Qualitative Results

Choose correct, difficult-but-correct, and failed cases from the galleries. Include both frozen-to-full improvements and regressions. For each case show the exact model input crop, ground truth, predictions and confidence, and an observation tied to the visible dish cues. Document the selection rule and use Grad-CAM panels where they help describe the model's focus.

Source material. `results/a2/error_analysis/qualitative_manifest.csv`, `results/a2/error_analysis/qualitative_review.csv`, galleries and Grad-CAM under `results/a2/figures/test/`.

11 Comparison and Discussion

Compare Table 1 with the per-class and compute evidence. Explain the mean paired gain of +0.1202 *macro-F1, increased full fine-tuning cost, and similar inference architectures for the two ResNet variants. Discuss the CNN's lower parameter count and compute cost alongside its accuracy. The CNN/ResNet comparison changes both architecture and initialization.*

Source material. `reports/a2/test_evaluation_analysis.md`, `results/a2/figures/test/a2_test_per_class_f1.png`.

12 Error Analysis

12.1 Error groups and examples

Build a table with group definition, error count and denominator, representative image IDs, plausible cause, and a future validation experiment. Start from the 525 errors in the validation-selected full seed-44 run. Show directed confusions such as Banh canh → Hu tieu (22/191) and Hu tieu → Banh canh (13/194). Explain that overlapping diagnostic groups may contain the same images.

12.2 Check the four EDA hypotheses

- H1 (rice-cake shape): report 13 within-pair confusions among 234 true Banh chung/Banh tet images.
- H2 (similar soup dishes): report 46 within-group confusions among 720 images from the three pre-specified soup classes.
- H3 (extreme aspect ratio): compare 4/52 with 521/4,932 and show crop examples. The observed rates do not support an elevated error rate in the extreme group.
- H4 (Banh pia background/packaging): explain the white-border proxy, the 0/11 vs. 2/78 counts, and the Grad-CAM observations. The shortcut hypothesis remains unresolved.

Source material. `results/a2/tables/a2_test_error_groups.csv`, `results/a2/error_analysis/hypothesis_results.json`, `reports/a2/test_evaluation_analysis.md`.

13 Ablation Study

Present frozen vs. full fine-tuning as the controlled factor study. State the hypothesis that backbone adaptation improves macro-F1, the shared data, augmentation, head learning rate, optimizer, epoch cap, patience and seeds, and the changes to backbone learning rate, gradients and BatchNorm behavior.

Report accuracy and primary macro-F1. Use the proposal criterion of a gain exceeding the observed spread across seeds. Interpret paired macro-F1 gains (+0.1150, +0.1236, +0.1219*) and the extra training cost.*

Source material. `reports/a2/proposal.md`, `results/a2/tables/a2_test_paired_seed_gains.csv`.

14 Limitations

Discuss web-image label ambiguity and source bias, residual near-duplicate risk, the single CNN draft run, and limited support for some error slices. Record that the official uncleaned test comparison in the proposal remains pending. Outline future validation studies for preprocessing and suspected background effects.

15 Conclusion

Answer the research question with the test metrics, paired macro-F1 gain and compute trade-off. Identify one unresolved failure pattern and a next experiment tied to it.

16 Reproducibility

Link installation and dataset setup, the split manifest, resolved configurations, all seven run IDs, seed values, checkpoint access, and evaluation commands. Trace each main result to its run, checkpoint/split/config hashes and source commit. Record software versions and hardware for training and inference.

Source material. Repository `README.md`, `configs/a2.yaml`, `results/a2/runs/`, `scripts/a2_evaluate_all.py`, `scripts/a2_error_analysis.py`, `scripts/a2_gradcam.py`.

AI Usage Disclosure

Reserved for the group's disclosure.

References

- CO3133 course handbook, Sections 3, 4, and 18–22.
- 30VNFoods dataset, Kaggle version 11. Add the dataset paper and dataset citation from the proposal.
- Add the ResNet paper and torchvision weight reference for the implementation used.
- Add the Grad-CAM paper cited in the test analysis.

A Dataset Proposal and Approval

Link the revised proposal and attach the confirmed approval record with any conditions.

Source material. `reports/a2/proposal.md`, `reports/a2/approval.md`.

B Supplementary Evidence

Include full per-seed tables and any supporting confusion matrices or qualitative cases that do not fit in the body. Map results to checkpoint and run identifiers.

Source material. `results/a2/tables/`, `results/a2/figures/test/`, `results/a2/runs/`.

C Team Contributions

Record each member's contributions after the group verifies them.